



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-032 Protected Practice Review

Diagnosis + changed retry prompts | open only after saved Pxx attempt

Artifact	C3_032_TRANSFORMATION_ENGINEERING_WAREHOUSING_REVIEW_PACK_PROTECTED_RC1
Status	PROTECTED - ATTEMPT REQUIRED
Phase	C3-02 Transformation Engineering & Warehousing 14 lessons 34-46 focused hours
Prerequisite	C3-01 Gate PASS. C2B warehouse/modeling foundations may be assumed; implementation depth is taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House rule: analogy -> tiny example -> guided real tool -> break/fix -> independent changed version -> validation
-> production reasoning -> evidence. A successful dbt command alone never proves correct data.



Review contract

This pack intentionally avoids a full copy-ready solved project. Compare your reasoning/control points, record the specific mistake, then close the review and complete a changed retry. Recognition is not mastery.





R01 - Layer boundaries

WHAT STRONG WORK CONTAINS

Raw stays reconstructable; each layer has a declared grain; business logic location and replay point are explicit.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Use support tickets; choose a different transformation boundary and defend it.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R02 - Determinism

WHAT STRONG WORK CONTAINS

Current-state winner needs business key + justified recency + tie-break; DISTINCT is not a repair.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Use price-history rows with same timestamp/different sequence.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R03 - dbt DAG

WHAT STRONG WORK CONTAINS

dbt parses project/Jinja and manages dependencies; DuckDB executes compiled SQL. source/ref define graph.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Add regions source->stg->dim and prove ref dependency.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R04 - Staging

WHAT STRONG WORK CONTAINS

Thin explicit names/types, source version grain retained, invalids visible.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Stage shipments with one bad quantity visibly.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R05 - Materialization

WHAT STRONG WORK CONTAINS

Choice follows size/freshness/cost/recovery; incremental adds state and must have rebuild path.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Choose for payments/current balance/monthly mart.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R06 - Tests

WHAT STRONG WORK CONTAINS

Tests map to grain/integrity; failure injection proves sensor; never weaken true failing test.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Build fresh returns test and controlled failure.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R07 - Incremental

WHAT STRONG WORK CONTAINS

Separate changed-row selection from target identity; base/delta/rerun/full-refresh evidence.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Inventory SKU update+insert, same-delta rerun.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R08 - History

WHAT STRONG WORK CONTAINS

Stable business key + change signal; only real changes create versions.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Subscription plan SCD history.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R09 - Macros

WHAT STRONG WORK CONTAINS

dbt/Jinja compiles to SQL; inspect compiled result; abstract repeated stable logic only.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Implement `safe_ratio` with zero denominator rule.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R10 - Docs/lineage

WHAT STRONG WORK CONTAINS

Docs state meaning/grain/keys/caveats; DAG shows dependency not business semantics.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Document refunds mart and lineage.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R11 - Dimensional integrity

WHAT STRONG WORK CONTAINS

Freeze fact grain before joins; one-side uniqueness; reconcile rows/keys/measures.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Shipment-item fact with duplicate carrier injection.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R12 - Late dimension

WHAT STRONG WORK CONTAINS

Preserve fact with controlled Unknown; keep source key; monitor/repair.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Claim with late provider P77.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R13 - Query plan

WHAT STRONG WORK CONTAINS

Correctness first; inspect scan/join/cardinality; optimization must preserve outputs.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Returns-by-product **EXPLAIN** + equivalent safe change.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.



R14 - CI/handoff

WHAT STRONG WORK CONTAINS

Clean reproducible checks must block; synthetic data; secret boundary; serving contract.

DIAGNOSE YOUR ATTEMPT

Write: (a) what you expected, (b) what actually happened, (c) the smallest wrong assumption/step, (d) which independent control exposed it or should have exposed it.

CHANGED RETRY - CLOSE THIS REVIEW FIRST

Daily returns PR-check/handoff.

MOVE-ON RULE

Changed retry passes its own controls and you can explain the failure/repair without this review open. Otherwise use targeted remediation for this competency only.